



QUANTITATIVE METHODS 1 · CHAPTER 3

Dispersion and Concentration

Detailed solutions

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1. Same mean, different risk

Team A daily sales are 48, 49, 50, 51, 52. Team B sales are 30, 40, 50, 60, 70. Compare their means and ranges. What does the comparison show?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. Both means equal 50. Ranges are $52-48=4$ and $70-30=40$. Interpretation. Equal averages do not imply equal stability: Team B is much more volatile. A measure of centre must therefore be accompanied by dispersion.

2. Variance and standard deviation

For $x=(2,4,4,6,9)$ calculate the population mean, deviations, squared deviations, variance and standard deviation.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. $\bar{x}=25/5=5$. Build the calculation table.

x_i	2	4	4	6	9
$x_i - \bar{x}$	-3	-1	-1	1	4
$(x_i - \bar{x})^2$	9	1	1	1	16

Result. Sum of squares =28; population variance $V=28/5=5.6$; standard deviation $=\sqrt{5.6}=2.37$. Interpretation. Observations typically lie about 2.37 units from the mean, although standard deviation is not a literal average distance.

3. Boxplot and outliers

For an ordered dataset, $Q_1=18$, median =24, $Q_3=30$, minimum =10, maximum =55. Compute the IQR and Tukey fences. Is 55 a potential outlier?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. $IQR=30-18=12$. Lower fence = $18-1.5(12)=0$; upper fence = $30+1.5(12)=48$. Since $55>48$, 55 is a potential upper outlier. Interpretation. "Potential" means the observation deserves investigation; it should not be deleted automatically.

4. Comparing relative dispersion

Store A has mean weekly sales EUR 20,000 and SD EUR 3,000. Store B has mean EUR 8,000 and SD EUR 1,600. Compare absolute and relative variability.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Use SD for absolute spread and $CV = s / \bar{x}$ for relative spread when means are positive and ratio-scale comparisons are meaningful. Result. A has the larger SD (3,000 versus 1,600). But $CV_A = 15\%$ and $CV_B = 20\%$. Interpretation. Store B is less variable in euros but more variable relative to its typical sales level.

5. Lorenz curve and Gini coefficient

Five households have incomes 10, 10, 20, 20 and 40. Construct cumulative population and income shares, then calculate the Gini coefficient using trapezoids.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Total income is 100. Population shares are $X=(0,.2,.4,.6,.8,1)$ and cumulative income shares are $Y=(0,.1,.2,.4,.6,1)$. The area under the Lorenz curve is $A=\sum (Y_{i-1}+Y_i)(X_i-X_{i-1})/2$. Result. The five trapezoids have areas .01, .03, .06, .10 and .16, so $A=.36$ and $G=1-2A=.28$. Interpretation. Inequality is positive but not extreme. The Gini summarises the whole distribution and should be reported with context or the Lorenz curve.

6. Dispersion from a frequency table

The number of weekly complaints per store is distributed as follows:

Complaints x	0	1	2	3	4
Stores n	4	8	10	6	2

- 1 Calculate the mean, population variance and standard deviation.
- 2 Calculate the coefficient of variation.
- 3 Interpret every result in context.

Method prompt. Build columns nx and nx^2 . Use $E(X^2) - [E(X)]^2$ for the population variance.

Detailed solution

Totals. $N = 30$; $\sum nx = 0+8+20+18+8 = 54$; $\sum nx^2 = 0+8+40+54+32 = 134$.

Mean. $54/30 = 1.80$ complaints.

Variance. $E(X^2) = 134/30 = 4.4667$. Therefore $\sigma^2 = 4.4667 - 1.80^2 = 1.2267$.

Standard deviation. $\sigma = \sqrt{1.2267} = 1.108$ complaints.

Coefficient of variation. $CV = 1.108/1.80 \times 100 = 61.6\%$.

Interpretation. Stores receive 1.8 complaints on average, with substantial heterogeneity relative to that mean. The CV is meaningful here because the ratio-scale variable has a genuine zero, but comparisons should use the same observation period and definition of a complaint.

7. Comparing concentration and robustness

Two markets have the following ordered firm market shares:

Market A	10%	15%	20%	25%	30%
Market B	2%	8%	10%	20%	60%

- 1 Construct cumulative shares for each market.
- 2 Explain which Lorenz curve is farther from equality.
- 3 For each market calculate the range and discuss why it is insufficient.
- 4 State a defensible concentration conclusion.

Method prompt. With five equal-weight firms, cumulative population shares increase by 20 percentage points.

Detailed solution

Cumulative shares. Market A: 0%, 10%, 25%, 45%, 70%, 100%. Market B: 0%, 2%, 10%, 20%, 40%, 100%. Population coordinates are 0%, 20%, 40%, 60%, 80%, 100%.

Lorenz comparison. At every interior population share, B's cumulative market share is lower. Its Lorenz curve is therefore farther below the equality line: B

Lorenz-dominates in the direction of greater concentration.

Ranges. A: $30 - 10 = 20$ points. B: $60 - 2 = 58$ points. The range confirms a difference but uses only two firms and ignores the middle of the distribution.

Conclusion. Market B is clearly more concentrated: the largest firm alone controls 60%, and the bottom 80% of firms jointly control only 40%. A complete report should show the Lorenz curve or a concentration index and should not infer market power from shares alone.